

ACCESS CONTROL

Card Reader

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Purpose

The purpose of this document is to define the standard sequence of operations for authorized personnel accessing a controlled door using an access control credential. This standard establishes consistent behavior for access control systems and supports security, auditing, and operational requirements.

This standard defines expected system behavior, communication requirements, monitoring requirements, alarm conditions, and operational expectations to ensure consistency across all physical security deployments.

Scope

This sequence applies to all badge-access controlled doors connected to the enterprise access control system.

This standard applies to:

- Access-controlled Doors
- Card Readers
- Access Control Panels
- Lockign Hardware
- Door Position Monitoring Devices
- Request-to-exit Devices
- Enterprise Access Control Management Systems

This standard applies to all new installations, upgrades, and retrofit projects.

Objectives

The objectives of this standard are:

- Ensure consistent user access experience
- Improve system reliability
- Support security monitoring
- Standardize deployment practices
- Improve troubleshooting capabilities
- Support long-term sustainment activities



System Components

An access control system is comprised of multiple hardware and software components working together to determine whether an individual is authorized to enter a secured area.

The following section provides an overview of each component and its role within the access control process.

Access Credential

An access credential is a physical or digital identifier issued to an authorized individual and used by the access control system to authenticate identity. The credential contains unique information that is evaluated against configured access permissions, schedules, and security policies to determine whether an individual is authorized to enter a secured area.

Credentials are currently issued to:

- Employees
- Contingents
- Vendors
- Visitors

For contingents, please refer to the correct personnel to issue a temporary badge:

- Associated Project Manager
- Badging Team – GSOC

OSDP Card Reader

A card reader is the device mounted near a controlled opening that receives credential information from a user.

The reader serves as the primary interface between the user and the access control system.

The current enterprise standard is Open Supervised Device Protocol (OSDP).

OSDP readers communicate with the access control panel using RS-485 communication and support:

- Two-way communication
- Device supervision
- Reader health monitoring
- Secure channel encryption



- Tamper monitoring

Access Control Panel

The access control panel is the decision-making component of the access control system. It acts as the intermediary between field devices and the access control management software.

The panel:

- Validates credentials
- Stores access permissions
- Controls door hardware
- Monitors alarms
- Generates events
- Maintains operation during temporary network outages

Electronic Locking Hardware

Electronic locking hardware physically secures the opening and prevents unauthorized access.

Common Types:

- Electric Strike
 - Unlocks the door frame strike allowing entry (fail-secure)
- Magnetic Lock (Maglock)
 - Uses electromagnetic force to secure the door (fail-safe)
- Electrified Mortise Lock
 - Provides integrated locking functionality within the door hardware

Door Position Switch (DPS)

A door position switch is a sensor used to monitor whether a door is open or closed. Also known as a door contact.

The sensor typically consists of:

- Reed Switch
- Magnet

Monitored Events:

- Door Open
- Door Closed
- Door Forced Open



Access Control

- Door Held Open

Request to Exit

A request-to-exit device allows authorized occupants to exit a secured area while maintaining security monitoring.

Common Types:

- Motion Sensor
 - Detects movement approaching the exit
- Push Button
 - Requires user activation
- Touch Sensor
 - Activated when touched

Access Control Management System

The access control management system is the software platform used to administer and monitor the physical security system.

Examples include:

- Lenel OnGuard
- Genetec Synergis
- CCURE 9000

The software is used to:

- Manage credentials
- Assign access levels
- Generate reports
- Monitor alarms
- Review audit logs
- Configure hardware

Network Infrastructure

The network infrastructure provides communication pathways between access control devices and management systems.

Components:

- Ethernet Switches
- Routers
- Firewalls



Access Control

- Servers
- Network Cabling

Reader Communication Standard

Enterprise Requirement

All newly installed readers shall utilize OSDP communication.

All currently installed Wiegand readers may remain operational until replacement or modernization projects occur.

OSDP vs Wiegand

Per the recent enterprise changes, OSDP is the new company standard for all card readers.

<i>Feature</i>	<i>Wiegand</i>	<i>OSDP</i>
Communication	One-Way	Two-Way
Encryption	No	Yes
Reader Supervision	No	Yes
Tamper Monitoring	Limited	Yes
Diagnostics	Limited	Advanced
Security Posture	Legacy	Enterprise Standard

OSDP Programming Requirements

The following requirements shall be implemented:

- **OSDP Secure Channel:** Enabled
- **Reader Supervision:** Enabled
- **Reader Tamper Monitoring:** Enabled
- **Unique Reader Address:** Assigned
 - **Format:** [CR-XX-XXXXXXX]
- **Reader Firmware:**
 - Please follow the firmware recommendation listed on *Reader Firmware Standards* document, found in the repository.



Normal Badge Access Sequence

Step 1 – Credential Presentation

The cardholder steps up to the badge reader and presents an authorized credential to the OSDP reader.

Expected Result:

- Reader receives credential information
- Reader transmits credential information to the access control panel

Step 2 – Credential Validation

The access control panel validates:

- Credential Status
- Credential Expiration
- User Access Level
- Time Schedule
- Anti-Passback Status (if enabled)

Expected Results:

- Access Granted
- Access Denied

Step 3 – Access Granted

Upon successful validation:

- Unlock door
- Generate Access Granted event
- Log transaction
- Start unlock timer

Unlock Time:

- 10 seconds

Step 4 – Door Monitoring

Door Position Switch (door contact) monitors opening activity.

System verifies:

- Door Opened



Access Control

- Door Closed
- Door Secured

Expected Result:

- Door returns to secure state after access transaction

Step 5 – Door Re-Source

Upon expiration of unlock timer:

- Lock re-engages
- Door secured
- Event logged

Access Denied Sequence

If the credential validation fails:

- Door remains locked
- Access Denied event generated
- Event logged

Common causes:

- Invalid credential
- Expired credential
- Inactive credential
- Unauthorized access level
- Access outside approved schedule

Door Forced Open Event

A Door Forced Open (DFO) event occurs when a secured door is opened without a valid access grant or authorized unlock command from the access control system.

Condition:

- Door opens without a valid access grant

System Actions:

- Generate DFO alarm
- Sound Local Noise
- Log Event
- Notify monitoring personnel (High Priority)



Door Held Open Event

Door Held Open (DHO) event shall be generated whenever a controlled opening remains open beyond the configured hold-open threshold following an authorized transaction or approved egress event.

Door Held Open thresholds shall be reviewed during design and commissioning activities to ensure alignment with operational requirements while maintaining appropriate security controls.

Condition:

- Door remains open longer than 15 seconds

System Actions:

- Generate Door Held Open alarm
- Sound Local Noise
- Log event
- Notify monitoring personnel (Medium Priority)

Request-to-Exit Sequence

Occupant activated approved request-to-exit device

System Actions:

- Unlock door
- Generate REX event
- Permit free egress

Reader Offline Event

Condition:

- Communication loss between reader and controller

System Actions:

- Generate Reader Offline event
- Notify maintenance personnel (Medium Priority)
 - Priority level changes based on Risk Level of area
- Log event

Audit Logging Requirements

The following events shall be retained:



Access Control

Event	Number of Days Retained
Access Granted	365
Access Denied	365
Door Forced Open	365
Door Held Open	365
Request-to-Exit	365
Reader Offline	365
Reader Online	365
Reader Tamper	365
Controller Offline	365
Controller Online	365

Risk Classification

Level	Risk	Description	Response
1	Critical	Life Safety Impact	Immediate
2	High	Security Vulnerability	24 hr
3	Medium	Operational Impact	30 days
4	Low	Documentation or Cosmetic	Next Scheduled Maintenance

Success Criteria

The access control system shall be considered compliant when:

- Authorized users gain access successfully
- Unauthorized users are denied access
- Events are logged correctly
- Alarms generate appropriately
- Readers remain supervised
- OSDP Secure Channel is enabled
- Door Monitoring functions operate correctly

Cadence

This standard shall be reviewed annually.

Lessons learned, operational findings, audit results, and technology changes shall be incorporated into future revisions to ensure continued effectiveness and program maturity.

Stakeholder feedback will be requested 3 months prior to future revision delivery dates.